

No.

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Ex. 4.4 | P120 | Ch. 4 | Chloe | 2025.7.30

例: (a) 构造 Y_0^0 $P_0(x) = \frac{1}{2^0 \cdot 0!} \frac{d^0}{dx^0} (x^2-1)^0 = 1$, $P_0'(x) = (-1)^0 (1-x^2)^{0/2} \frac{d^0}{dx^0} P_0(x) = 1$

$$Y_0^0(\theta, \phi) = \sqrt{\frac{(2 \cdot 0 + 1)}{2\pi} \cdot \frac{(0-0)!}{(0+0)!}} e^{i \cdot 0 \cdot \phi} \cdot P_0^0(\cos \theta) = \frac{1}{\sqrt{4\pi}}$$

构造 Y_2^1 $P_2(x) = \frac{1}{2^2 \cdot 2!} \frac{d^2}{dx^2} (x^2-1)^2 = \frac{1}{8} (12x^2-4) = \frac{1}{2} (3x^2-1)$

$$P_2'(x) = (-1)^1 (1-x^2)^{1/2} \frac{d^1}{dx^1} P_2(x) = -3x \sqrt{1-x^2}, \quad P_2'(\theta, \phi) = \sqrt{\frac{(2 \cdot 2 + 1)}{4\pi} \cdot \frac{(2-1)!}{(2+1)!}} e^{i \cdot 1 \cdot \phi}$$

$$P_2'(\cos \theta) = -\sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{i\phi}$$

(b) 验证正交归一性

- Y_0^0 归一性 ($l=l'=0, m=m'=0$)

$$\int_0^{2\pi} \int_0^\pi |Y_0^0|^2 \sin \theta d\theta d\phi = \int_0^{2\pi} d\phi \int_0^\pi \sin \theta \cdot \frac{1}{4\pi} d\theta = \frac{1}{4\pi} \cdot 2\pi \cdot 2 = 1, \text{ 归一。}$$

- Y_2^1 归一性 ($l=l'=2, m=m'=1$)

$$\int_0^{2\pi} \int_0^\pi |Y_2^1|^2 \sin \theta d\theta d\phi = \frac{15}{8\pi} \int_0^{2\pi} d\phi \int_0^\pi \cos^2 \theta \sin^3 \theta d\theta = \frac{15}{8\pi} \cdot 2\pi \cdot \frac{4}{15} = 1, \text{ 归一。}$$

- Y_0^0, Y_2^1 正交性 ($l \neq l'$)

$$\int_0^{2\pi} \int_0^\pi (Y_0^0)^* Y_2^1 \sin \theta d\theta d\phi = \int_0^{2\pi} \int_0^\pi \frac{1}{\sqrt{4\pi}} \cdot \frac{15}{8\pi} \cdot \cos^2 \theta \sin^3 \theta d\theta = 0 \text{ (奇)}$$

故正交。